

# *Week 17*

Keel Bone Chicken Project

12/22/2025



College of Engineering

# ***This Week***

## **This week's progress**

- **PCB Reordering**
  - Purchase Request sent to Procurement Office
- **PCB Design**
  - Updates to Rev. 4 of PCB Design
  - Added Force + Strain Logging
  - Several Design Choices to Minimize Area
- **Data Analysis**
  - Able to determine chicken movement from PCB
  - Need to change some firmware on the Nicla Sense

# ***PCB Reordering***

Purchase Request sent to Procurement Office

- Drill files regenerated and issue dealt with with JLCPCB support
- Need to call procurement office to ensure the order successfully goes through
- May be an issue with the resistors
  - 2 mm (length) x 1 mm (height) sizing too small for regular PCBA
  - May need to switch to 4 mm (length) x 2 mm (height) for our PCB

# ***PCB Design***

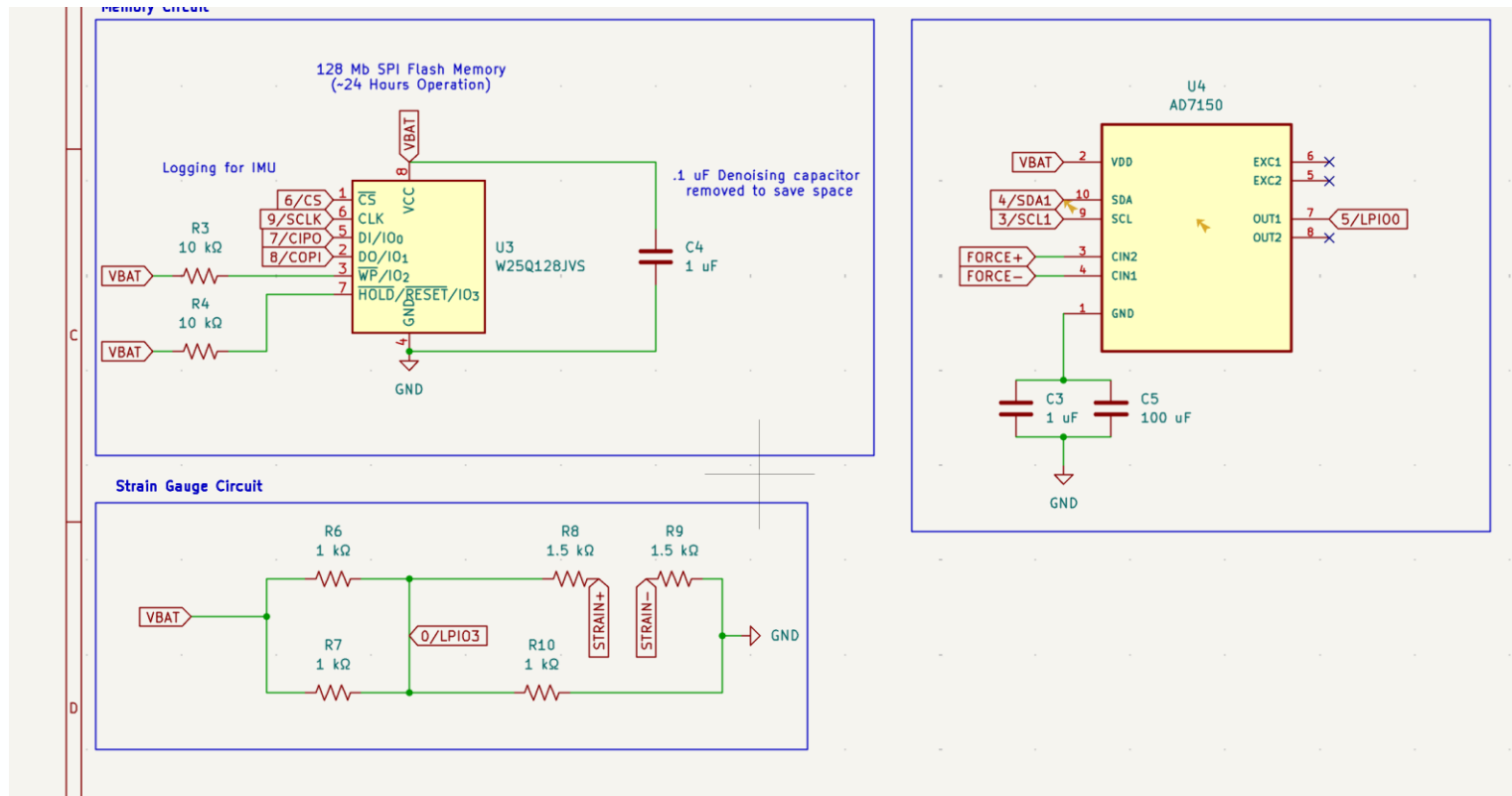
Updated Rev. 4 of PCB Design

- Design Files + Footprints found and generated
- Footprint of added hardware is smaller than anticipated, which is good
- Minimal Issues with Routing this time around

# PCB Design

## Added Force + Strain Logging

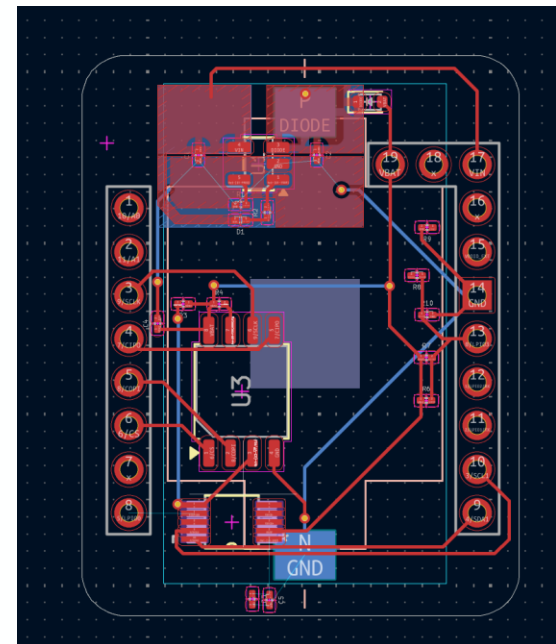
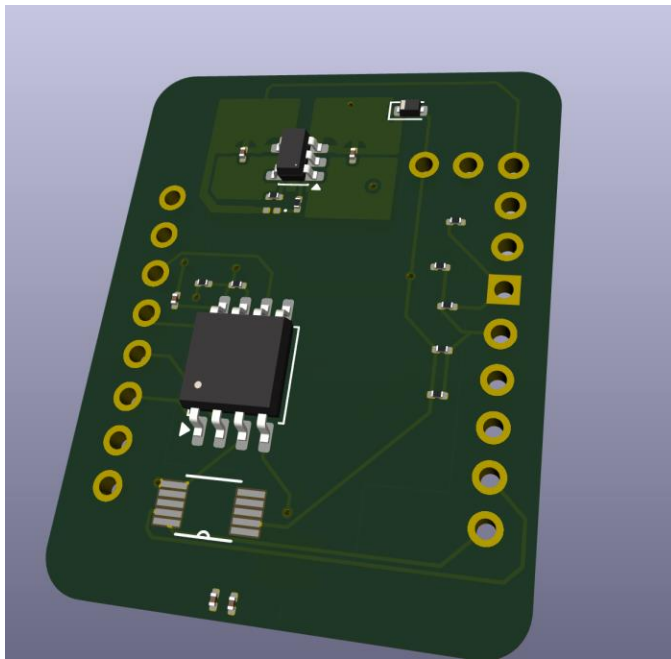
- Sensor Circuitry can be seen in the Schematic below...
- Added to PCB on the next slide



# PCB Design

## Several Design Choices to Minimize Area

- Second Smallest Resistors Chosen due to Assembly Issues
- Took precautions to minimize taken area
  - Space needed for Ribbon/Flat Flex Cables in the Future
  - Space needed for power dissipation in the future



# Data Analysis

Able to determine chicken movement from PCB

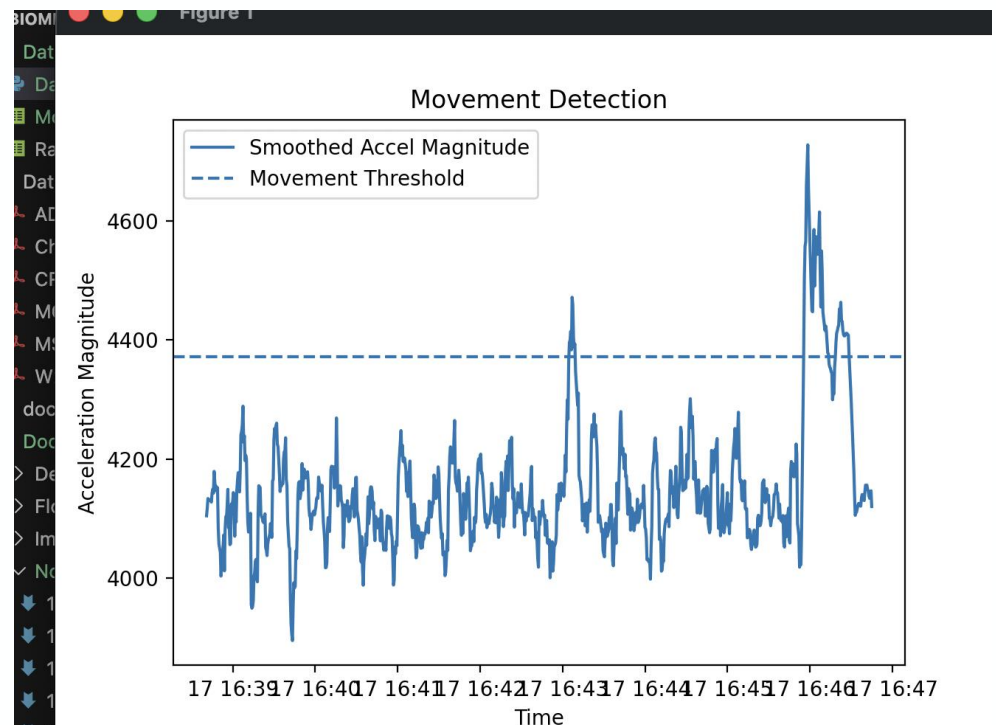
- From the PCB, ~16 minutes 37 seconds of data was logged
  - I had originally set the maximum entries to 1000 -> 16:40
  - Missing 3 seconds of data is good, likely startup time
  - From the data, I was able to run the following data smoothing program:

```
Data > DataProcessing.py > ...
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4
5 PATH_NAME = "Movement1.csv"
6
7 df = pd.read_csv(PATH_NAME)
8
9 df["timestamp"] = pd.to_datetime(df["timestamp"])
10
11 df["accel_mag"] = np.sqrt(
12     df["accelerometer_Ax"]**2 +
13     df["accelerometer_Ay"]**2 +
14     df["accelerometer_Az"]**2
15 )
16
17 # Rolling Window
18 df["accel_mag_smooth"] = df["accel_mag"].rolling(
19     window=10, center=True
20 ).mean()
21
22 # Movement threshold
23 threshold = [
24     df["accel_mag_smooth"].mean() +
25     2 * df["accel_mag_smooth"].std()
```

# Data Analysis

Need to change some firmware on the Nicla Sense

- IMU seems to turn off after extended periods of time on battery (20+ minutes)
  - Need to look into this issue and see how it can be avoided in the future
  - From data collection, I was able to obtain the following plot
  - Very visible when the chicken choose to move vs. fly





# *Next Week*

## Blockers & Next Week's Work

- Need to find the suitable battery replacement on the PCB
- Work on Hybrid Approach for Data Collection
- Fix Issue where Nicla IMU turned off unexpectedly after a prolonged time of data collection
- Instruct Agriculture Department (Omosalewa) on what to do with the boards

# *Thank You*

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